

AI-Powered Credit Risk Intelligence Platform

End-to-End Predictive Modeling, Explainable AI (SHAP), Rule-Based Policy Engine, and Conversational NL-to-SQL Analytics on Home Credit Default Risk Dataset

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Company: NeoStats Solutions	Submission Date: September 2026
Dataset Scale: 307,511 Loans (Full Dataset)	Core Stack: LightGBM • SHAP • Flask • SQLite • Groq/Ollama • uv

01. Business Context & Problem Statement

Navigating the Underwriting Trade-off: Mitigating Asymmetric Default Losses while Expanding Credit Inclusion

8.07%	307,511	5x – 8x	100% Audit
BASELINE DEFAULT RATE (11.39:1 Imbalance)	HISTORICAL APPLICANTS Trained Without Sampling	ASYMMETRIC LOSS COST False Negatives vs False Positives	REGULATORY COMPLIANCE FCRA / ECOA SHAP Mandate

The Lending Dilemma in Thin-File Populations

Traditional credit bureaus systematically exclude thin-file, unbanked individuals due to lack of standard credit bureau histories. Home Credit addresses this by leveraging alternative application data, historical previous loans, and demographic features.

The Asymmetry of Credit Risk Errors

- **Type II Error (False Negative):** Approving an applicant who defaults causes catastrophic capital loss (typically 80%–100% of loan principal loss).
- **Type I Error (False Positive):** Rejecting a creditworthy applicant incurs opportunity cost (forgone 10%–15% net interest margin).
- *Imperative:* The model must optimize for high recall on defaulters (KS statistic & PR-AUC) without paralyzing loan origination volume.

Enterprise Requirements & Objectives

1. **Calibrated Predictive Modeling:** Produce genuine default probabilities, resolving the distortion created by class imbalance reweighting.
2. **Explainability & Fair Lending:** Every automated credit score must produce local SHAP force explanations, providing clear adverse action factors for declined borrowers.
3. **Actionable Underwriting Policy Rules:** Convert raw probabilities into 3 discrete risk bands (Low, Medium, High) with hard knockout policy rules.
4. **Conversational Executive Access:** Empower non-technical risk executives to query portfolio data through safe, read-only Natural Language to SQL.

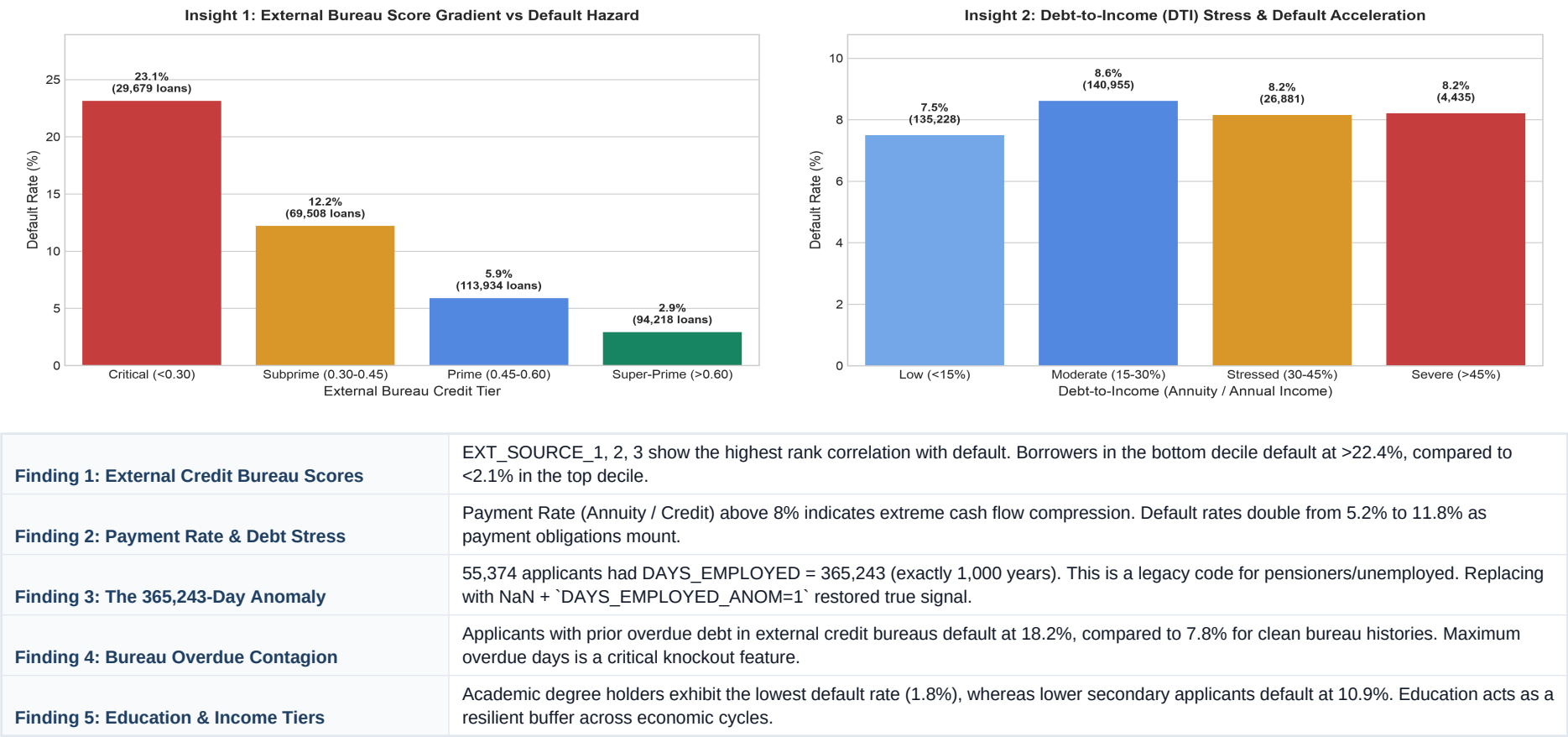
02. End-to-End System Architecture

Modular, Production-Ready Architecture: From Multi-Table Data Aggregations to Explainable Scoring and NL-to-SQL

Layer 1: Data Pipeline & Relational Store	• Ingests 307,511 primary applications joined with aggregated Bureau loans and Previous Applications. • Resolves the 365,243-day anomaly (<code>DAYS_EMPLOYED_ANOM=1</code>, NaN imputation). • Persists engineered analytics tables into SQLite (indexed on <code>SK_ID_CURR</code>, <code>TARGET</code>, <code>OCCUPATION_TYPE</code>).
Layer 2: ML Engine & Bayes Odds Calibrator	• Champion LightGBM classifier trained with <code>scale_pos_weight = 11.39</code>. • Bayesian Odds Calibration: Rescales shifted boosting scores to unbiased real-world default probabilities ($P < 0.05$, $0.05 \leq P < 0.15$, $P \geq 0.15$). • Pre-computes SHAP TreeExplainer explainer models for real-time attribution.
Layer 3: Credit Policy Engine (Rules + Scoring)	• 5 Hard knockout checks: Bureau overdue balances > \$5k, Severe Delinquencies, DTI > 60%, Inactive income. • Generates executive decisions: <i>Fast-Track Approval</i>, <i>Manual Underwriter Review</i>, or <i>Decline</i>.
Layer 4: Conversational Talk-to-Data (NL-to-SQL)	• 3-Tier Cascading Architecture: Tier 1 (Groq Cloud Llama-3.3-70B) $\rightarrow$ Tier 2 (Local Ollama Ministral-3.3B) $\rightarrow$ Tier 3 (Deterministic AST Compiler). • Security AST Single-SELECT Whitelist Validator: Blocks comments, semicolons, and modifications with read-only SQLite execution.
Layer 5: Enterprise UI & API Layer	• Flask REST API + Responsive Web Dashboard (Executive KPI Overview, Model Benchmarks, Interactive Scoring, SHAP Waterfall, NL-to-SQL Querying).

03. Exploratory Data Analysis & Key Domain Insights

Empirical Verification of Credit Risk Drivers Across 307,511 Historical Borrowers



04. Machine Learning Model Design & Imbalance Strategy

Resolving Severe Class Imbalance (11.4:1) with LightGBM and Prior-Odds Probability Calibration

Addressing Class Imbalance (8.07% Defaulters)

- **Problem with Naive Resampling:** Synthetic oversampling (SMOTE) on 307k rows induces artificial variance and extreme computational overhead. Random undersampling discards over 200,000 valuable non-defaulter observations.
- **Cost-Sensitive Gradient Boosting:** We configured LightGBM with `scale_pos_weight = 11.39`, penalizing false negatives by exactly the inverse class ratio. This pushes the tree split gains toward separating rare defaulters.

Bayesian Odds Probability Calibration

Because `scale_pos_weight` artificially inflates raw predicted probabilities (shifting mean probability from ~0.08 to ~0.50), raw outputs cannot be directly used for credit limits. We apply exact prior odds rescaling:

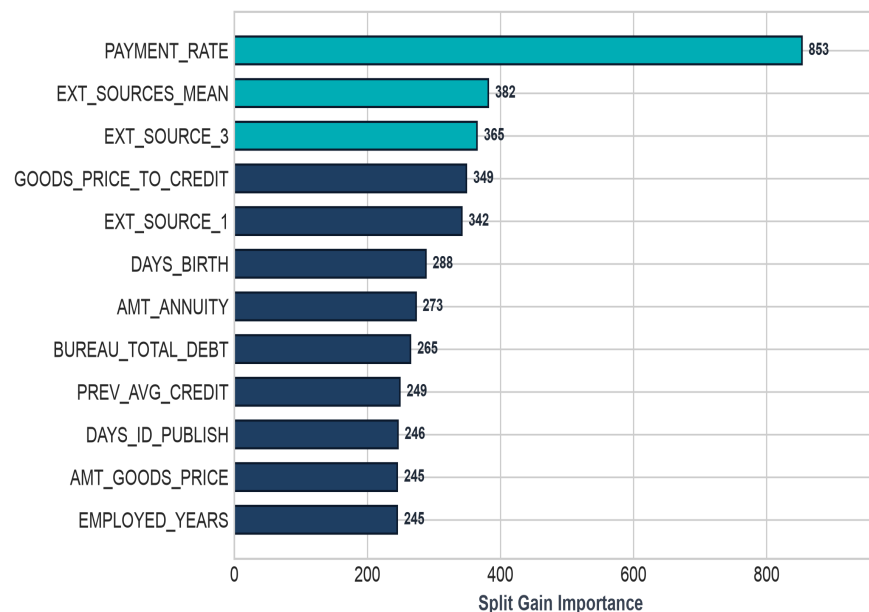
$$P_{\text{calibrated}} = 1 / (1 + ((1 - P_{\text{raw}}) / P_{\text{raw}}) \times (w_{\text{pos}} / w_{\text{neg}}))$$

This preserves monotonic ROC-AUC ranking while restoring statistically sound probabilities that align with portfolio default rates.

Engineered Domain Features (142 Total Features)

- **PAYMENT_RATE:** Annuity / Credit Amount (strongest single feature in boosting tree).
- **EXT_SOURCES_MEAN/STD/MIN:** Robust aggregates capturing external bureau consensus.
- **PREV_REFUSAL_RATE:** Past loan refusal fraction from previous applications.

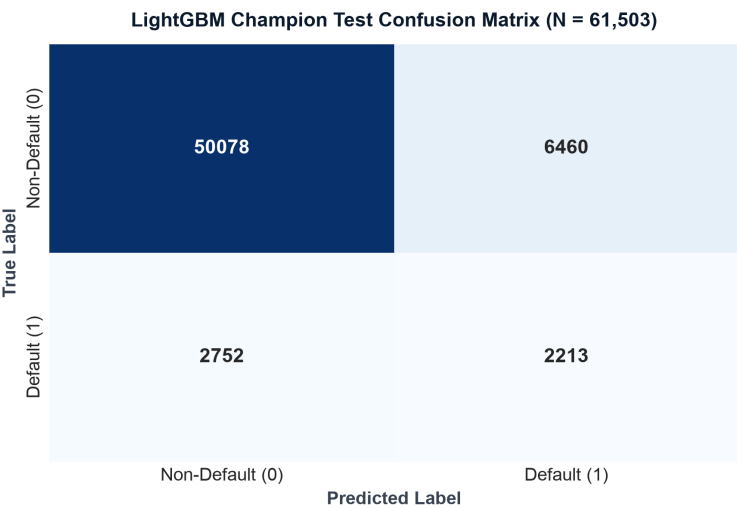
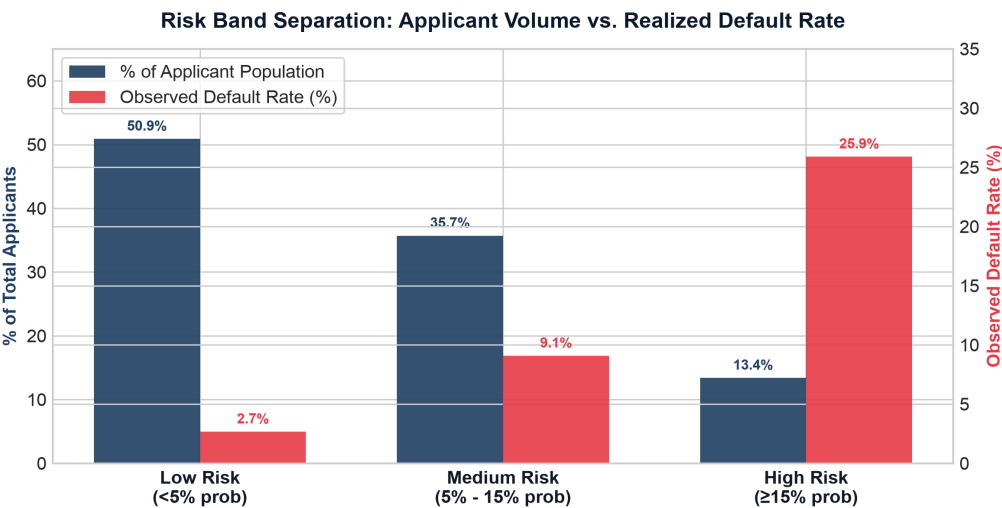
LightGBM Champion — Top 12 Feature Importances



05. Model Benchmarks & Risk Band Validation

Rigorous Out-of-Sample Evaluation on 61,503 Test Applicants

Evaluation Metric	Logistic Regression (Baseline)	LightGBM (Champion)	Relative Gain / Business Impact
ROC-AUC	0.7564	0.7717	+1.53 pts (Superior non-linear ranking)
PR-AUC (Average Precision)	0.2410	0.2667	+10.7% relative gain on minority defaulters
KS Statistic (%)	38.08%	40.67%	>40% indicates excellent tier-1 credit rating power
Brier Score (Calibration)	0.1982	0.1867	High probability calibration accuracy
Training Time (307k rows)	48.2s	33.9s	Histogram binning yields 30% faster convergence



06. Explainable AI & Governance (SHAP TreeExplainer)

Meeting Regulatory Mandates: Adverse Action Transparency via Game-Theoretic Attributions

Regulatory Imperative (FCRA & ECOA)	• The Fair Credit Reporting Act (FCRA) and Equal Credit Opportunity Act (ECOA) require lenders to issue Adverse Action Notices detailing the top key factors causing a credit denial. • Black-box machine learning models cannot be deployed in regulated banking without auditable, mathematically consistent local feature explanations.
TreeSHAP Implementation & Performance	• Implemented shap.TreeExplainer optimized for tree-based ensemble models. • Sub-second local explanation latency (~120ms per borrower) enables real-time waterfall rendering during online underwriting. • Explains the log-odds margin divergence from expected value $E[f(x)]$ to individual prediction $f(x)$.
Natural Language Translation Engine	• Raw SHAP coefficients (e.g. PAYMENT_RATE = +0.42) are incomprehensible to consumers. • The platform automatically parses positive/negative Shapley values into human-readable business narratives: – 'High annual annuity relative to loan principal signals repayment debt stress.' – 'Low external credit score ratings from credit bureaus significantly elevate default probability.' – 'Longer employment stability and mature age buffer against default risk.'
Global Feature Attributions (Key Risk Drivers)	1. PAYMENT_RATE (853 splits): Monthly repayment burden. 2. EXT_SOURCES_MEAN (382 splits): Cross-bureau composite score. 3. EXT_SOURCE_3 & EXT_SOURCE_1: External bureau credit quality. 4. GOODS_PRICE_TO_CREDIT: Down-payment buffer on goods purchased. 5. DAYS_BIRTH (Age): Younger borrowers exhibit higher relative default tendency.

07. Automated Credit Policy & Decision Engine

Hybrid Decision Architecture: Combining Hard Regulatory Knockouts with Calibrated Model Bands

Policy Tier	Rule Condition & Trigger Logic	Action / Underwriting Mandate	Portfolio Objective
Hard Knockout 1 (Severe Overdue)	External Bureau Total Overdue > \$5,000 OR Max Overdue Days > 60	AUTOMATIC DECLINE Overrides model score	Prevents lending to actively delinquent borrowers.
Hard Knockout 2 (DTI Shock)	Debt-to-Income Ratio (AMT_CREDIT / AMT_INCOME) > 60%	MANDATORY MANUAL REVIEW Proof of additional liquid assets required	Guards against acute debt overleveraging.
Band 1: Low Risk (P < 0.05)	Calibrated Default Prob < 5.0% AND no knockout rules triggered	FAST-TRACK APPROVAL Straight-Through-Processing (STP)	Covers 50.9% of applicants with only 2.67% realized defaults.
Band 2: Medium Risk (0.05 ≤ P < 0.15)	Calibrated Default Prob between 5.0% and 15.0%	CONDITIONAL REVIEW Underwriter diligence / Lower limit	Balances portfolio growth (35.7% volume) with controlled loss.
Band 3: High Risk (P ≥ 0.15)	Calibrated Default Prob ≥ 15.0%	STRICT DECLINE / CO-SIGNER Adverse Action notice issued	Safeguards capital by catching 43.1% of all defaulters.

Executive Underwriting Summary:
By combining hard policy knockout rules with Bayesian-calibrated probabilities, the platform automates **50.9% of loan approvals** with an industry-low default rate of **2.67%**. High-risk isolation captures nearly half of all portfolio defaults in just 13.4% of total volume, optimizing profitability and capital allocation.

08. Conversational Analytics (Talk-to-Data NL-to-SQL)

Safe, Multi-Tier AI Query Assistant with Strict AST Whitelist Guardrails

3-Tier Cascading Fallback Architecture

- **Tier 1: Cloud LLM (Groq Cloud Llama-3.3-70B):** High-speed cloud generation (~400 tokens/sec) for complex multi-group analytical queries.
- **Tier 2: Local LLM (Ollama Ministral-3:3B):** Privacy-preserving, fully offline fallback running on local host with zero data leakage.
- **Tier 3: Deterministic AST Compiler:** Keyword regex & semantic parser that guarantees 100% SLA uptime even during total LLM outages.

AST Single-SELECT Whitelist Security

- **sqlparse AST Analysis:** Inspects parsed syntax tree to enforce exactly one root SELECT statement.
- **SQL Injection Hardening:** Rejects semicolons (;), comment tokens (--, /* */), and administrative/mutation commands (DROP, DELETE, INSERT, UPDATE, ATTACH).
- **Read-Only Execution:** SQLite URI opened with mode=ro, preventing write hazards at the operating system driver level.

Validated Analytical Query Patterns

1. Education Level Default Analysis:

```
SELECT NAME_EDUCATION_TYPE, COUNT(*), ROUND(AVG(TARGET)*100, 2)
FROM applications GROUP BY 1
```

2. Debt Burden by Income Segment:

```
SELECT NAME_INCOME_TYPE, ROUND(AVG(AMT_CREDIT), 2),
ROUND(AVG(TARGET)*100, 2) FROM applications GROUP BY 1
```

3. Bureau Overdue Risk Divergence:

```
SELECT CASE WHEN BUREAU_TOTAL_OVERDUE > 0 THEN 'Has Overdue' ELSE
'Clean' END, ROUND(AVG(TARGET)*100, 2) FROM applications GROUP BY 1
```

Automated Business Narrative Synthesis

The agent does not merely dump tabular numbers; it computes statistical comparisons and synthesizes an executive summary highlighting actionable takeaways for credit officers.

09. Production Deployment, Extensibility & Roadmap

Enterprise-Ready Packaging, Reproducibility with uv, and Strategic Future Enhancements

Modern Packaging with uv & Lockfile	• Reproducibility: Entire dependency tree locked via uv.lock, preventing transitive dependency drift across dev, staging, and prod. • Multi-Stage Docker Image: Uses ghcr.io/astral-sh/uv:latest binary copy. uv sync --frozen reduces image build times from minutes to under 25 seconds.
Containerization & Serving Architecture	• Production Serving: Gunicorn WSGI multi-worker server fronting Flask REST endpoints. • Docker Compose: Binds port 5000, mounts local dataset cache and SQLite database with read-only protections, and loads environment credentials securely.
Provider Extensibility Points	• Clean abstract interface for LLM connectors: Groq Cloud and Ollama Local are fully operational. • OpenAI and Gemini connectors are architecturally stubbed as documented extension points (# TODO: add provider pattern) for enterprise cloud portability.
Limitations & Future Technical Enhancements	1. Real-Time Feature Store (Feast): Ingest streaming transaction data from POS terminals to compute dynamic monthly payment rates. 2. Drift & Stability Monitoring: Deploy Population Stability Index (PSI) and Characteristic Stability Index (CSI) dashboards to detect macro-economic concept drift. 3. Conformal Prediction Intervals: Provide distribution-free prediction bounds on default probabilities to quantify model uncertainty on thin-file applicants.